

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

Master of Science in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

Bachelor of Science in Mathematics, Minor in Computer Science | GPA: 3.74/4

Austin, TX

Aug. 2023 – May 2026

Relevant Courses: Data Structure, Stochastic Process, Elements of Software Design, Data Analytics, Principles of Database System, Discrete Math, Real Analysis, Probability and Statistic, Artificial Intelligence, Elements of Networking

WORK EXPERIENCE

Enterprise Risk Analyst Intern

Toyota Industries Commercial Finance, Inc.

May 2025 – Dec 2025

Dallas, TX

- Developed a full-stack credit risk modeling pipeline (SQL Server → Azure) with ETL, feature engineering (e.g., PayNet inquiries, LTV ratios) and ML models (Logistic Regression, Random Forest), improving default prediction accuracy by 3%.
- Validated and backtested models across asset segments (Class 8 trucks, trailers, used vehicles), achieving AUC > 0.82 and revealing 3.7× higher default rates in high-risk groups, driving refinements to Toyota's credit scorecard logic.
- Automated monthly model refreshes via parameterized SQL + Python workflows integrated into Power BI dashboards, reducing manual reporting time by 10% and enabling real-time risk tracking.

Financial Data Development Engineer – Fraud Detection

Caitong Securities Ltd

Jul. 2024 – Aug. 2024

Hangzhou, China

- Designed and optimized SQL-driven ETL pipelines processing 10M+ daily transactions, using window functions, subqueries, and stored procedures to improve extraction efficiency by 5%.
- Implemented automated detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% and improving reliability.
- Developed a customer risk scoring engine by integrating multiple transactional and behavioral data streams, enhancing fraud detection precision and supporting AML compliance.

Software Engineering Intern – Customer Analytics

Joblogic-X Corporation

Nov. 2023 – Jun. 2024

Dallas, TX (Hybrid)

- Designed and implemented a scalable data pipeline integrating social media and transactional data for customer analytics, supporting data-driven marketing and engagement strategies.
- Developed ranking and clustering algorithms in Python, integrating a KOC scoring module into an AI-powered chatbot via REST API to enhance automation and personalization.
- Applied modular design and Git-based version control to ensure reproducibility, maintainability, and scalability across cross-platform deployment environments.

Data Scientist Assistant Intern – Image Restoration

Software Engineering Center, Chinese Academy of Sciences

Jul. 2023 – Sep. 2023

Beijing, China

- Deployed and optimized image restoration modules (VAE loss, Gaussian filters, LSGAN) on cloud servers via SSH, improving workflow stability and scalability for large-scale image reconstruction.
- Integrated GFPGAN API for automated face restoration and color correction, and containerized the system with Git for reproducible, cross-platform deployment and performance benchmarking.

RESEARCH EXPERIENCE / PROJECTS

CodeAssist Project

University of Texas at Austin | Mentor: Prof. Kia Teymourian | Tools: Flask, Docker

Aug. 2025 – Present

- Developed modular backend architecture to integrate multiple AI providers (OpenAI, Hugging Face) via API routing, reducing vendor dependency and optimizing performance-cost tradeoffs.
- Implemented dynamic model-switching logic for large-scale deployments, improving scalability, reliability, and adaptability across multi-cloud environments.
- Debugged and enhanced front-end components in React, fixing feature update issues such as the “updated list” and “course description” display to ensure seamless UI-backend integration.
- Collaborated on system refactoring and Docker containerization to improve maintainability, reproducibility, and cross-team development efficiency.

Researcher & TA — Biological Data Analytics

University of Texas at Austin | Mentor: Prof. Becca Young | Tools: R, Linux

Jan. 2024 – Dec. 2024

- Processed and analyzed 250GB of RNA-seq data using Linux, Docker, and AWS Cloud, optimizing computational workflows to improve processing speed by 6% and reduce storage costs by 5%.
- Applied bioinformatics pipelines (Salmon, RSEM, Trinity, samtools) for transcript quantification, achieving a 98.7% mapping rate and ensuring reproducibility across multiple datasets.
- Performed differential expression analysis with Voom + Limma, identifying 48 key genes across time-series samples and applying permutation-based FDR to enhance statistical robustness.

- Automated R-based workflows for normalization, PCA visualization, and volcano plot generation using ggplot2, improving interpretation efficiency by 15%.
- Collaborated with peers as a teaching assistant to train incoming FRI students on RNA-seq pipelines, promoting reproducibility and data literacy in early-stage biological research.

FinChat — AI-Powered Financial Analysis System

Southern Methodist University | Mentor: Prof. Labiba Jahan | Tools: Python

Jan. 2023 – May 2024

- Collaborated with Prof. Labiba Jahan and a cross-university team to build FinChat, an AI-driven platform for automating financial report analysis through large language models (LLMs).
- Developed ETL pipelines to extract and process 100+ SEC filings from EDGAR using the FinGPT API, structuring unstructured financial data into a scalable data lake for metric extraction and trend detection.
- Implemented and optimized LangChain-based LLM workflows integrated with CrewAI for contextual financial interpretation, improving data processing accuracy by 3% and reducing manual analysis time.

Kiwifruit Quality Grading Project — Machine Learning Member

Huazhong Agricultural University | Mentor: Prof. Yan Tong | Tools: Python

Aug. 2022 – May 2024

- Assisted in developing an ML pipeline for automated fruit grading using multi-feature extraction (color, texture, shape) and fusion techniques; contributed to model benchmarking and experimental analysis for the manuscript.
- Performed comparative evaluation of feature combinations to optimize grading accuracy and interpret results for inclusion in the paper's ML methodology section, published in *Expert Systems with Applications (Elsevier)*.
- Explored distributed system and blockchain integration for kiwifruit traceability, enhancing data reliability and transparency

Machine Learning Research Assistant — Neuroengineering Lab

Remote | Mentor: Dr. Kunqiang Qing | Tools: Python, MATLAB

Jan. 2023 – May 2024

- Processed and analyzed 35GB of multimodal EEG/fNIRS data using MATLAB and Python (Pandas, NumPy, MNE) for signal cleaning, visualization, and connectivity analysis in brain-computer interface research.
- Applied statistical testing (ANOVA, post hoc) and feature extraction methods to evaluate cognitive patterns, contributing to two peer-reviewed papers published in *Frontiers in Neurology* and *Frontiers in Neuroscience*.
- Collaborated on the design of emotion recognition models using classical ML algorithms (RF, SVM, KNN, XGBoost), strengthening understanding of experimental design and academic writing in applied machine learning.

LEADERSHIP & COMMUNITY INVOLVEMENT

Leader — Austin Chinese Campus Christian Fellowship

Aug. 2023 – June 2026

University of Texas at Austin

- Organized weekly Bible study sessions, retreats, and cultural activities such as hiking trips and traditional Chinese festival celebrations, fostering spiritual growth and cultural connection among members.
- Coordinated logistics for transportation and event scheduling, improving group communication and ensuring inclusivity for members with diverse backgrounds and schedules.
- Developed leadership, communication, and empathy by mentoring new members, facilitating group discussions, and resolving interpersonal challenges with patience and understanding.

Volunteer — Art From the Streets

Austin, TX

- Supported community-based art programming for artists experiencing homelessness or housing insecurity, helping create a welcoming and respectful environment for participants.
- Assisted with event preparation, artwork organization, and visitor engagement, contributing to accessible public art experiences and community outreach.
- Strengthened communication, empathy, and service-oriented leadership through direct involvement with artists, volunteers, and local community members.

PUBLICATIONS

- 1 Miao, H., Wang, Z., **Zhang, Y.**,... & Tong, Y. (2025). New kiwifruit quality grading methods based on image multi-feature fusion and ResTNet model. *Expert Systems with Applications*, 129507. (IF = 7.5)
- 2 Xu, M., **Zhang, Y.**, Zhang, Y, Liu, X., & Qing, K. (2024). EEG biomarkers analysis in different cognitive impairment after stroke: *Frontiers in Neurology*, 15, 1358167. (IF = 2.8)
- 3 **Zhang, Y.**, Zhang, Y, Jiang, Z., Xu, M., & Qing, K. (2023). The effect of EEG and fNIRS in digital therapy of Alzheimer's disease: *Frontiers in Neuroscience*, 17, 1269359. (IF = 3.2)
- 4 **Zhang, Y.** (2023, August). EEG-Based on Emotion Recognition Using Machine Learning. In *IEEE ICIPCA* (pp. 226–232). IEEE.
- 5 Li, D., Zhang, Z., & **Zhang, Y.** (2023). Public compassion under the mediation of suffering: Taking the videos of "Second Uncle" on social media as an example. *Media and Communication Research*, 4(5), 46–58.